



ESBM 3700 ENTREPRENEURIAL ENVIRONMENTS

LECTURE 10

CHOOSING YOUR TECHNOLOGY



University of Colorado **Boulder**

ENTREPRENEURIAL STRATEGY: THESE CHOICES MATTER



CHOOSE YOUR
STRATEGY



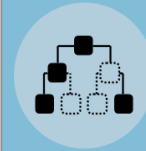
CHOOSE YOUR
CUSTOMERS



CHOOSE YOUR
TECHNOLOGY



CHOOSE YOUR
COMPETITION



CHOOSE YOUR
ORGANIZATION



VALUE CREATION HYPOTHESIS



VALUE CAPTURE HYPOTHESIS

Our path

Module 1

Entrepreneurial Mindset

Ideas to Opportunities

Business Models

Sources of Funding

Lean Startup

Lean Startup Exercise

Venture Pitching

Module 2

Choosing Your Customer

Choosing Your Technology

Choosing Your Identity

Choosing Your Competition

Quiz

Module 3

Disruption Strategy

Intellectual Property Strategy

Value Chain Strategy

Architectural Strategy

Bonus: Startup Failure

Bonus: Shark Tank

Final Exam

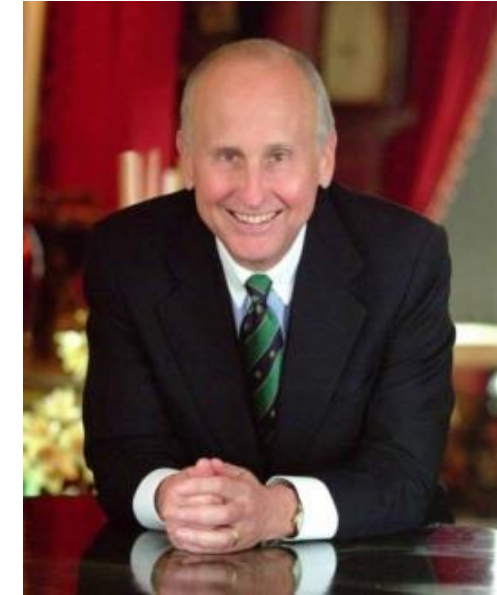
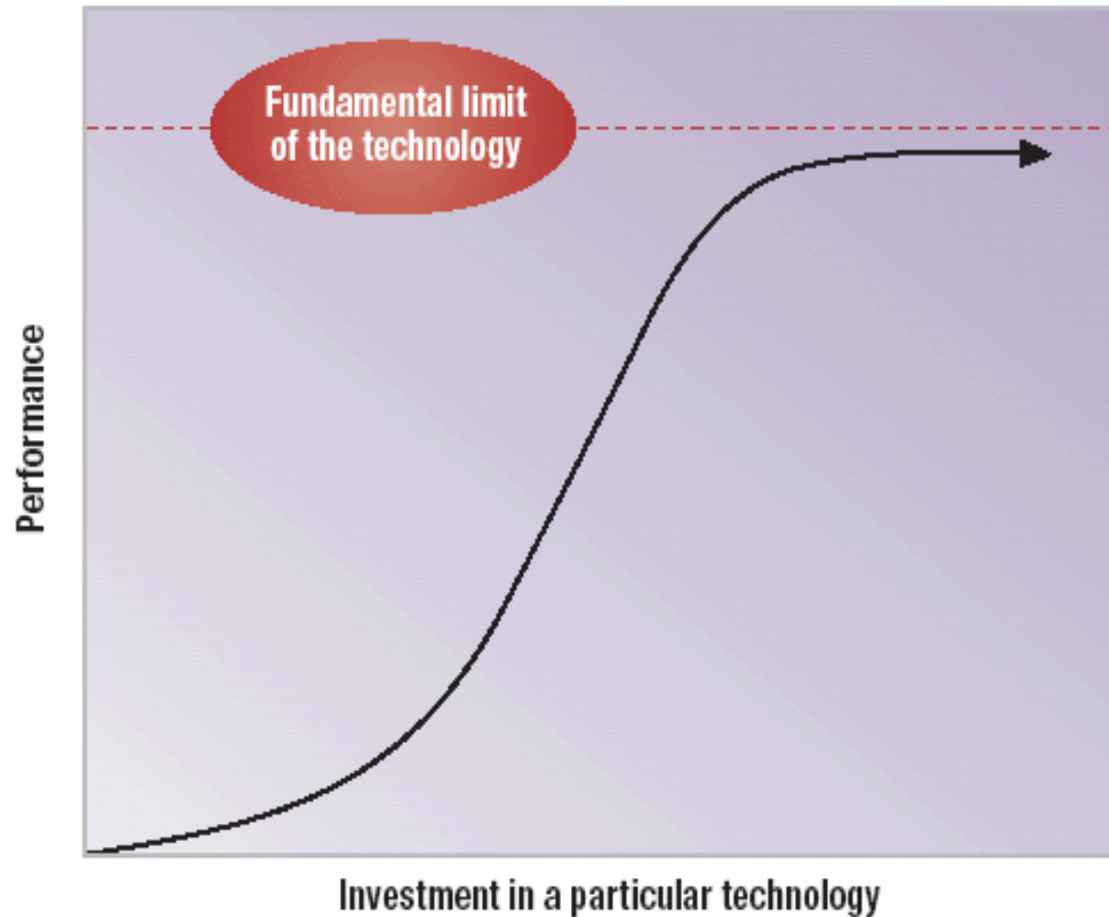
Technologies for an application



- Which technology should be selected for autonomous driving?

<https://www.youtube.com/watch?v=ojJz3SGoQWM>

The Technology S-Curve



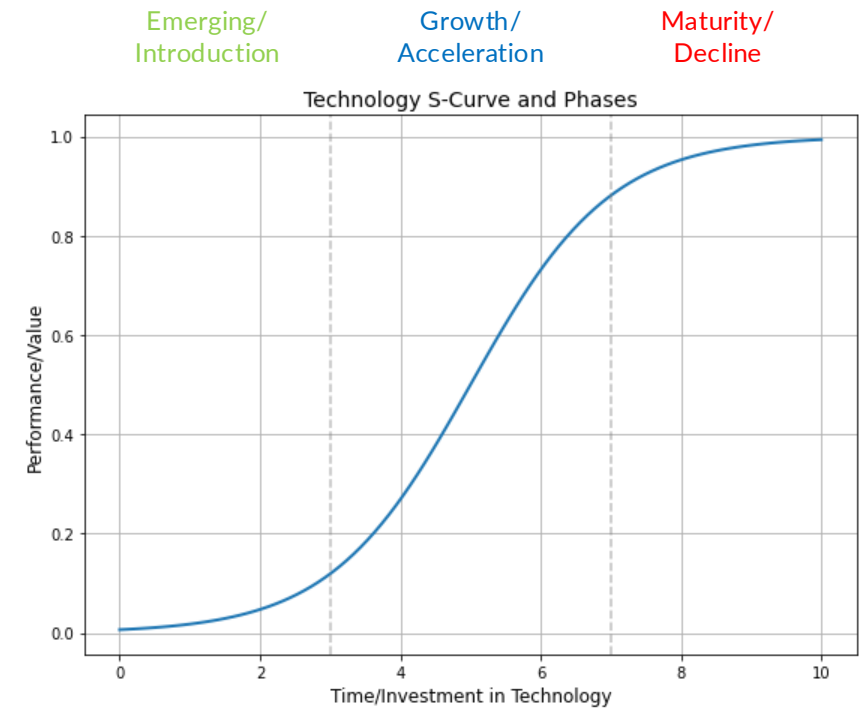
Richard Foster
proposed the notion of
technology S-curves while
in McKensie & Co

The Technology S-Curve

- The Technology S-Curve illustrates the **life cycle of a technology**, showing how performance improvements typically slow over time as the technology matures.

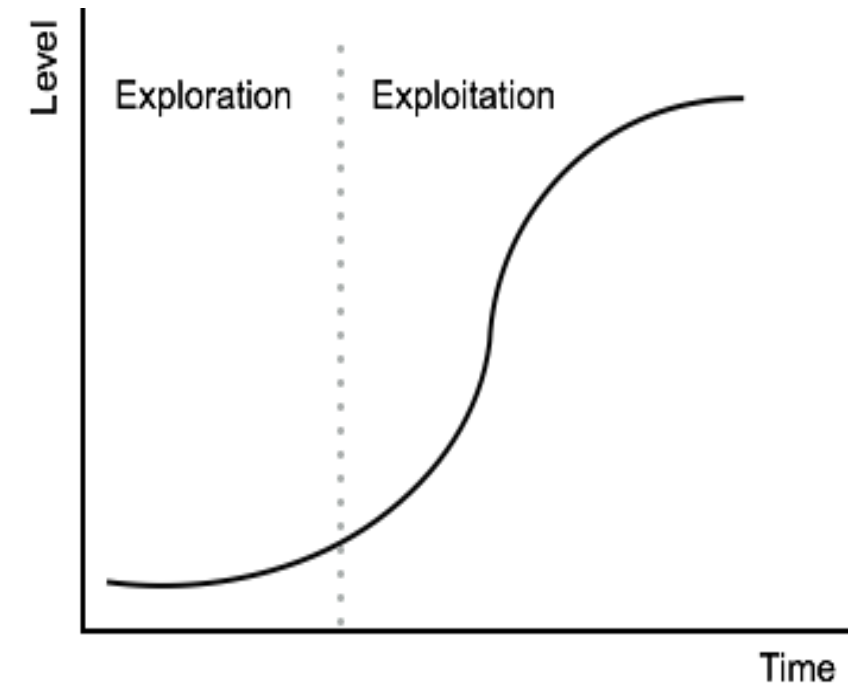
Key Phases:

- **Emerging/Introduction:** In the early stages, there is slow progress as firms experiment with the technology. High uncertainty and investment in R&D are common, and returns are initially low.
- **Growth/Acceleration:** Once the technology reaches a tipping point, its performance improves rapidly, leading to exponential growth in adoption and value.
- **Maturity/Decline:** As the technology matures, the rate of improvement slows down, and returns diminish. Eventually, the technology may become obsolete or replaced by a new, disruptive innovation.



The Technology S-Curve

- **Exploration:** Exploration refers to the pursuit of new knowledge, innovation, and experimentation with emerging technologies. It involves searching for new opportunities, developing novel products or processes, and pushing the boundaries of what is possible.
 - **Characteristics:**
 - High uncertainty and risk.
 - Investment in R&D and radical innovation.
 - Focus on long-term gains and potential breakthroughs.
- **Exploitation:** Exploitation involves leveraging existing technologies, knowledge, and capabilities to optimize processes, improve efficiency, and maximize short-term performance. It focuses on incremental improvements and refining current operations.
 - **Characteristics:**
 - Low risk and high predictability.
 - Emphasis on efficiency and optimization.
 - Focus on short-term gains and performance improvement.



Exploration in cell phones

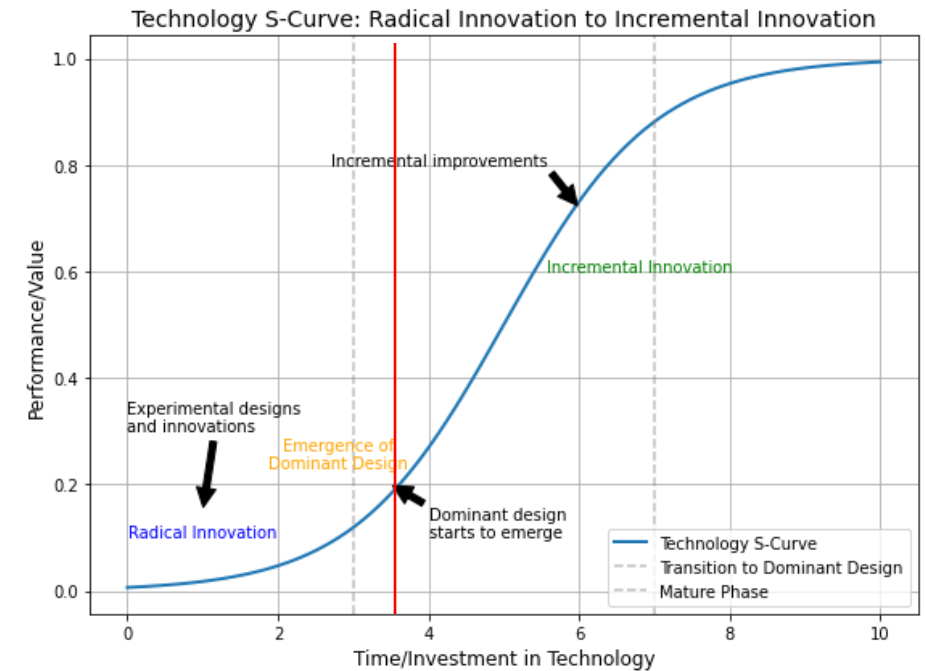


Exploitation in cell phones



The Technology S-Curve

- **Dominant Design:** The dominant design is a **product architecture** or **technology standard** that emerges after a period of experimentation and competition between multiple alternatives. Once a dominant design is established, it typically becomes the benchmark, and subsequent innovations focus on refining and optimizing that design.
- The emergence of a dominant design signals the end of **radical innovation** and the beginning of **incremental innovation**. Firms often shift from exploring new designs to exploiting the established design for greater efficiency and performance.

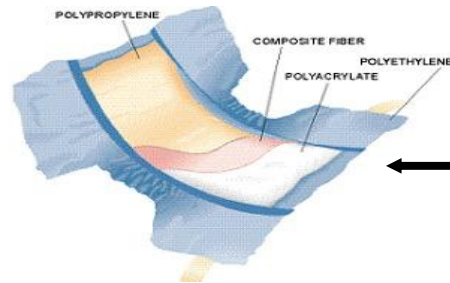


Smaller: Evolution of the disposable diaper

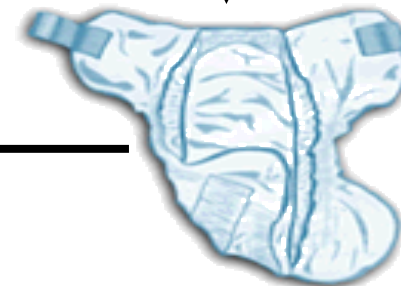
The 1960s – Vic Mills' Pampers!



The 1940s – Marion Donovan's "The Boater"

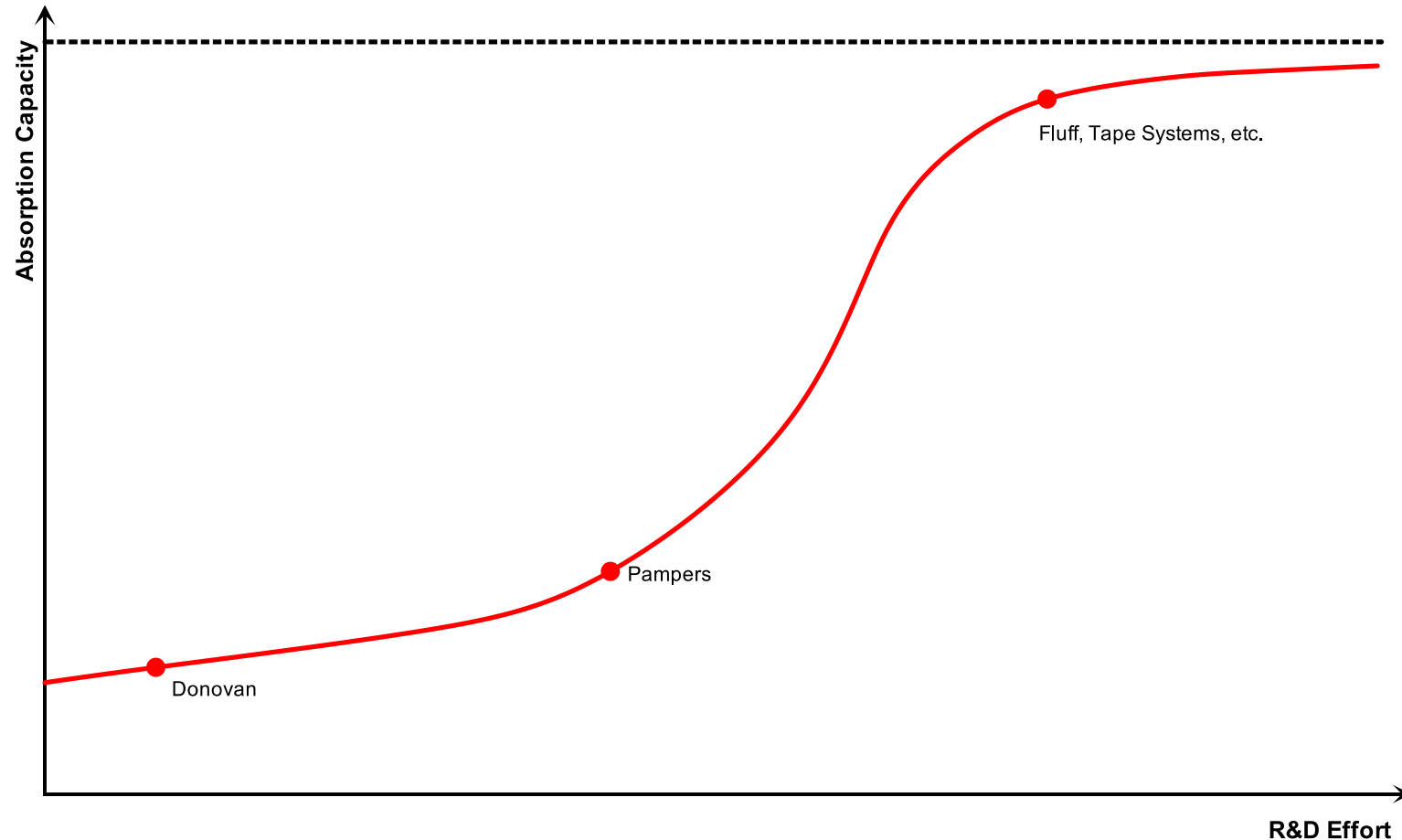


The 2000s – Another 60% *smaller*!

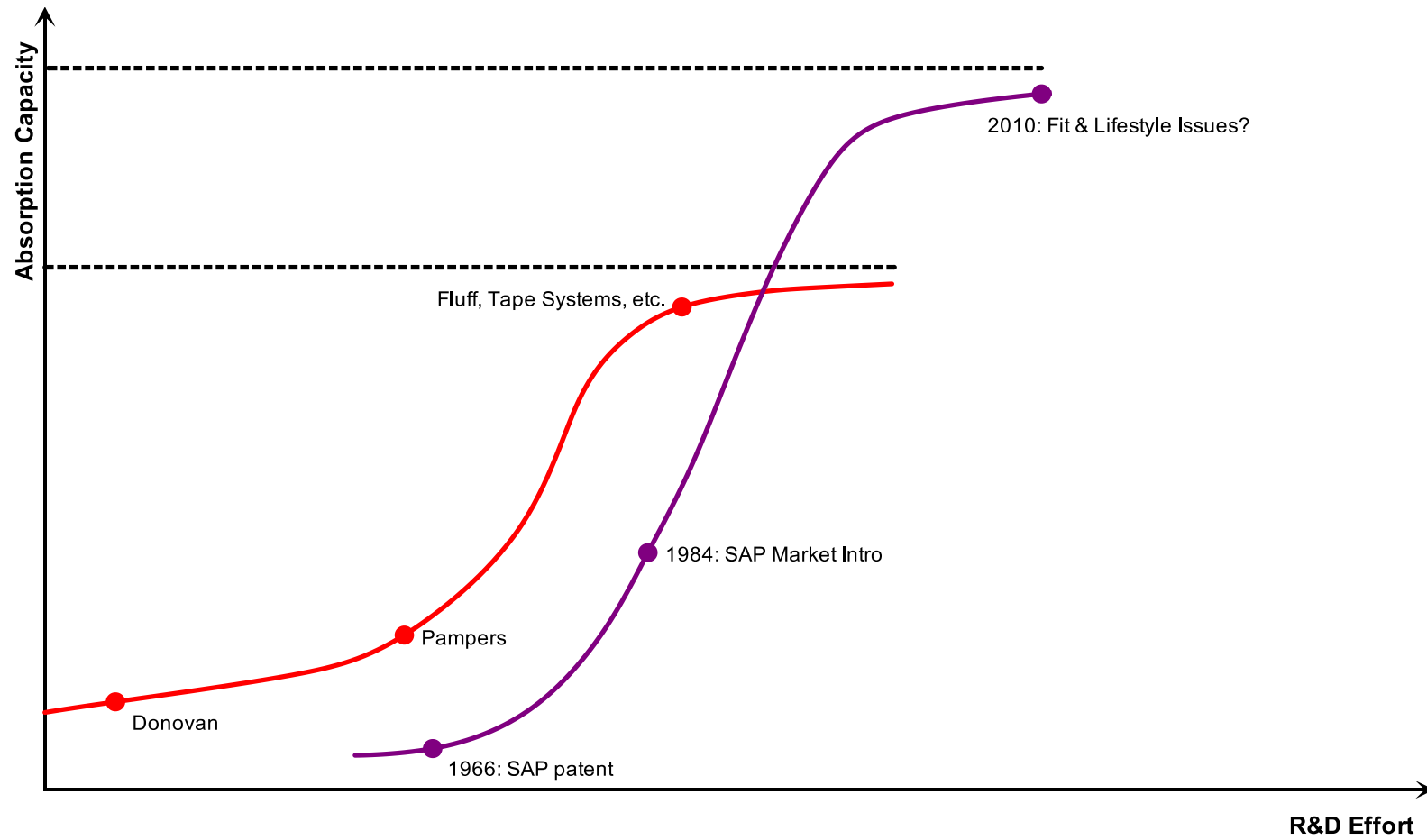


The 1980s – Harper and Harmon
Super-Absorbent Polymer

The absorbent core diaper faced significant technological hurdles due to the tradeoff between size and absorbency...

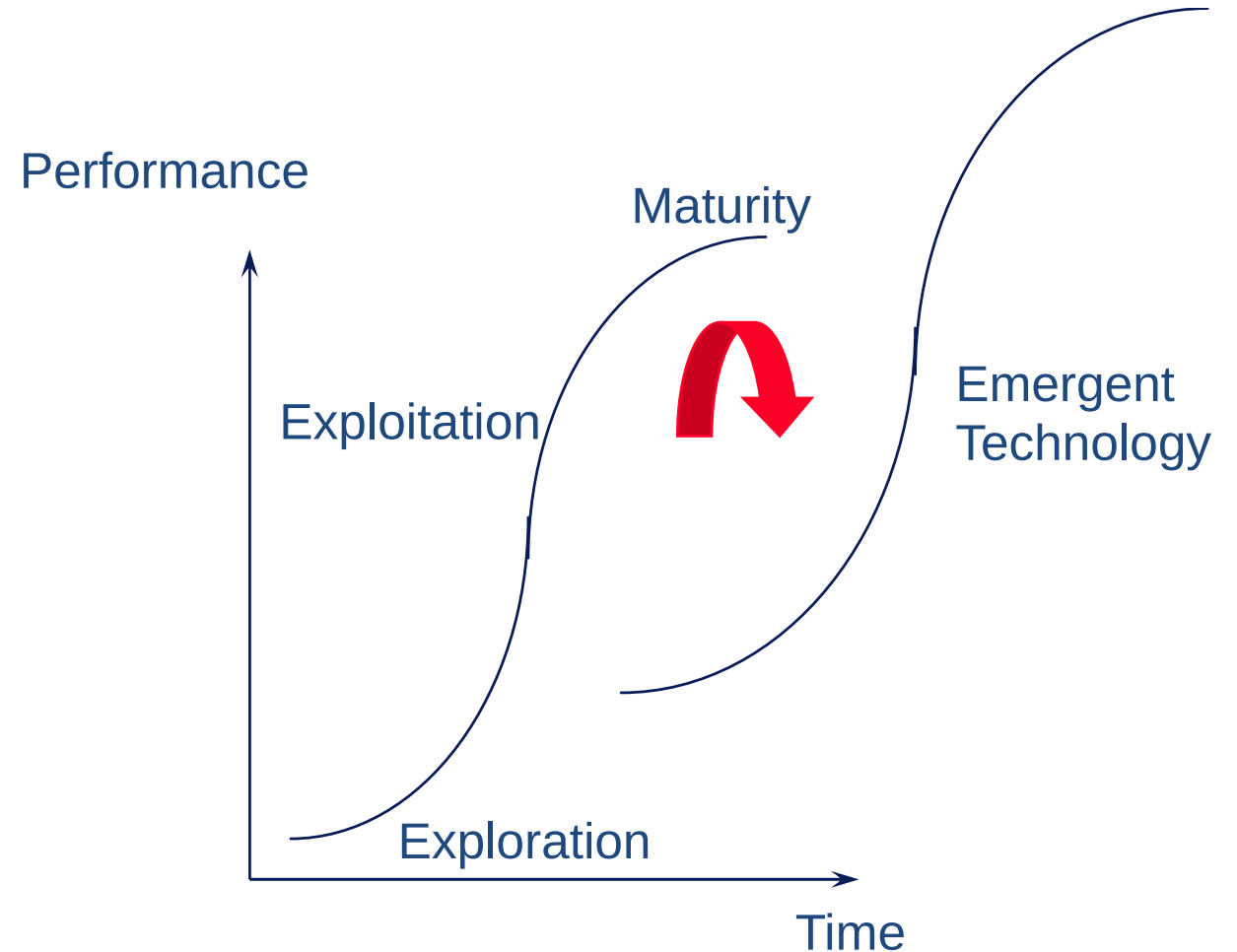


However, the SAP technology facilitated a long period of sustainable improvements in absorption capacity, with benefits for fit, storage, transport/logistics, cost, and the environment!

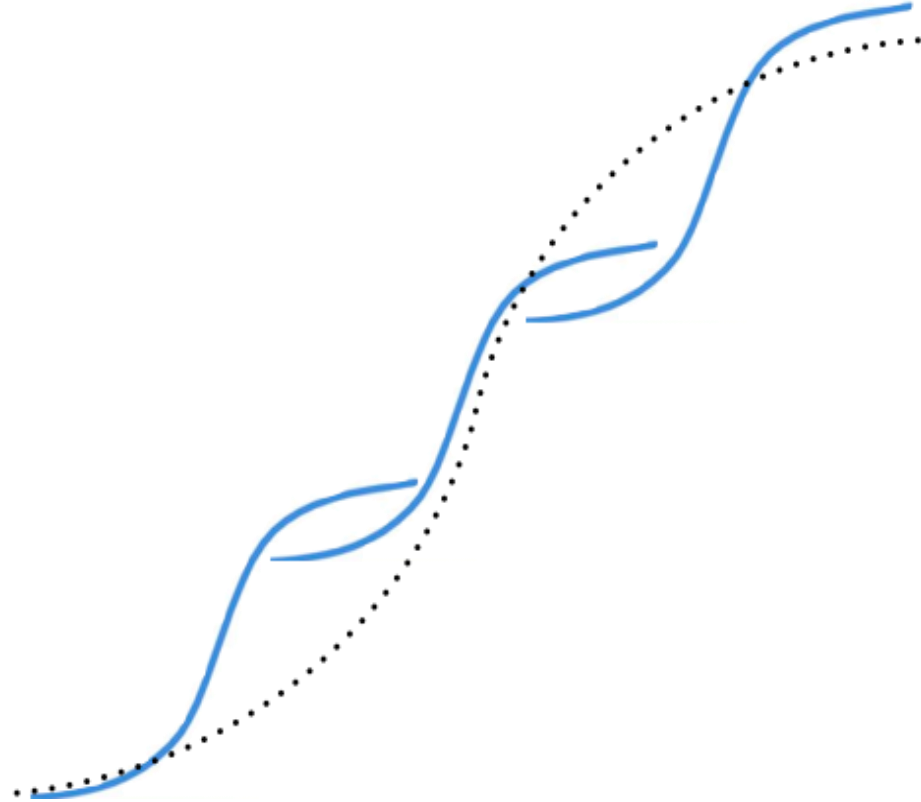


The Technology S-Curve at Work

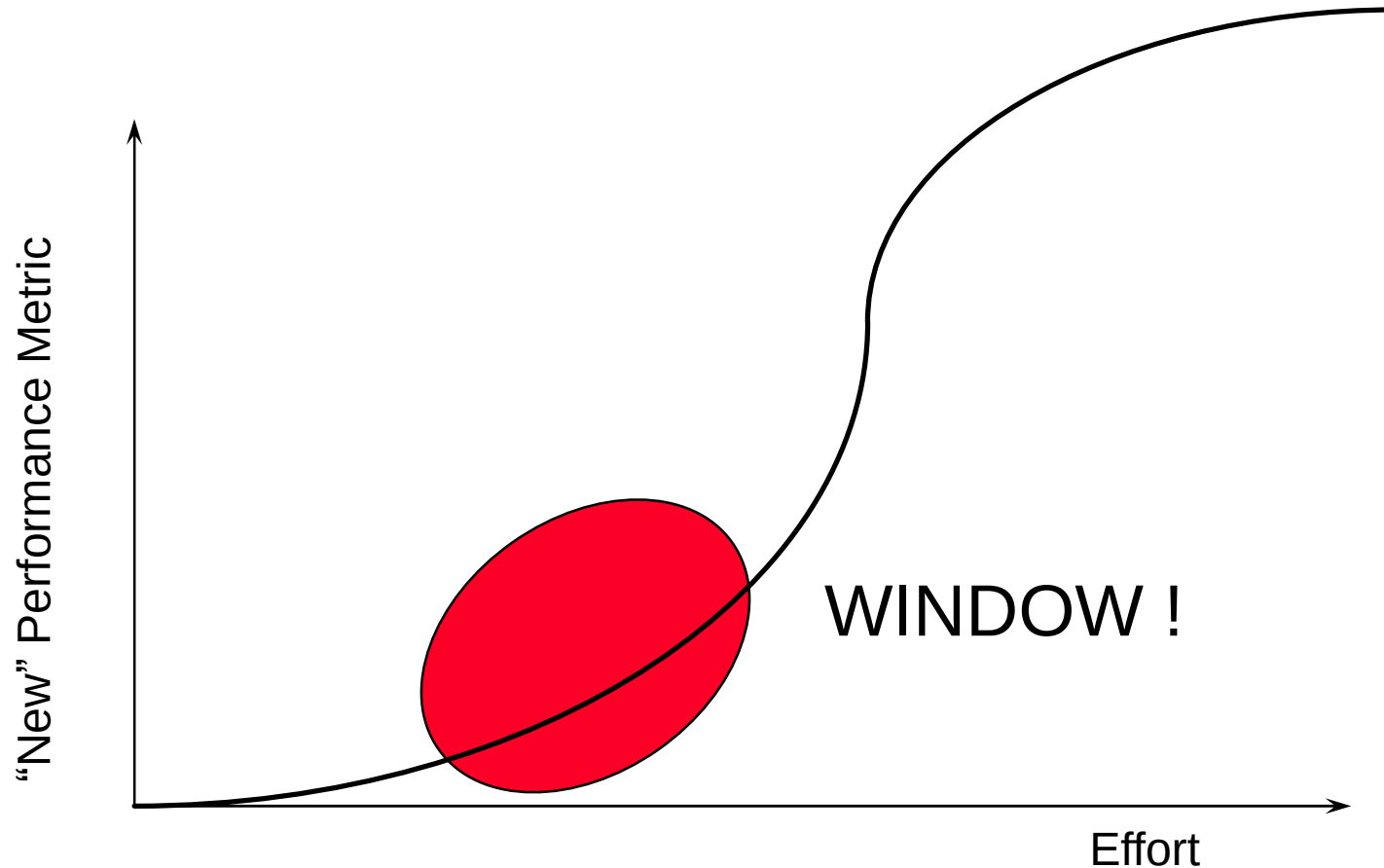
- Technology disruption occurs when a **new**, often **inferior** or **experimental**, technology emerges and eventually replaces or displaces an existing dominant technology.
- The new technology follows its own S-curve, starting small but eventually surpassing the existing one in terms of performance, value, or market share.



**Similar to the Market S-Curve,
the Technology S-Curve is “fractal,” made of a
numerous more micro S-curves....**



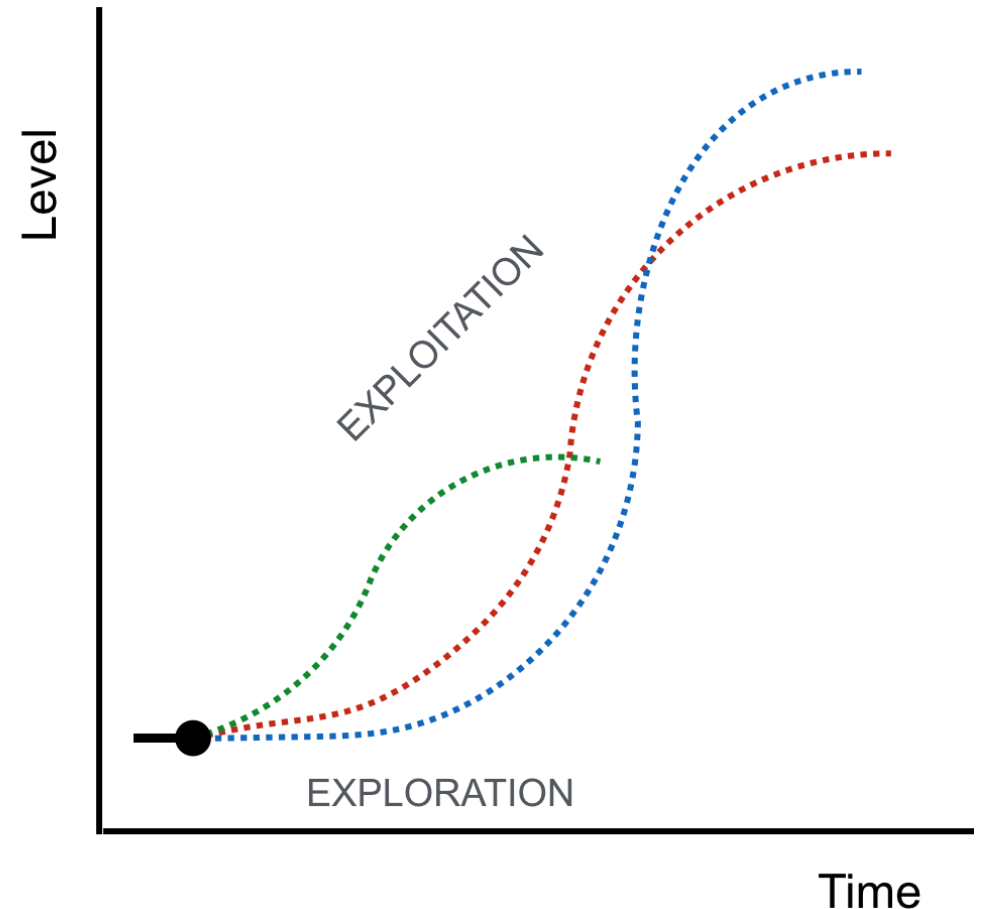
The “Window of Opportunity”



For a start-up, new S-curves allow you to choose a technology to reshape an industry.

Choosing Your Technology

- An entrepreneur can choose:
 - When to jump on the technology S-curve, and
 - Which technology S-curve to jump on
- **Short S-curves** have a brief phase of growth before reaching maturity.
 - **Advantages:** Quick returns on investment, fast time to market, but with low long-term growth.
 - **Risk:** May be easily disrupted or rendered obsolete by newer, more innovative technologies.
- **Tall S-curves** have a longer growth phase, with sustained improvements over time.
 - **Advantages:** Long-term market dominance, significant returns on investment, and the ability to fend off competition for longer.
 - **Risk:** Takes longer to realize the full potential and may require higher initial investment in R&D.



The Technology S-Curve at Work

- The Technology S-Curve is a powerful and widely used tool for tools for the analysis of technology evolution, but it is *extremely* important to think carefully about the technologies you are studying and adapting the framework to fit the application.
- The framework can be applied at multiple “levels” of analysis, and it is possible to gain insights from playing around with different levels of “aggregation”.
- As a forward-looking tool, S-curve analysis helps start a discussion about what kinds of capabilities to invest in, what types of problems to focus on, and the potential for emerging technologies to overturn existing sources of competitive advantage.

THANK YOU!